

BRIEF

The Cryptographic Cliff: Preparing for Q-Day

Post-quantum migration timelines for DoD.

EXECUTIVE SUMMARY

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Post-quantum migration timelines for DoD. This brief distills IST's research and convening across government, industry, and the investment community into findings that decision-makers can act on now.

The analysis draws on structured interviews with program officials and industry engineers, a review of budget and acquisition data, and workshops held under the auspices of the Institute's Quantum computing portfolio. Three themes recur: the pacing challenge posed by near-peer competitors, the mismatch between authorities and timelines inside the Department, and the underused leverage of private capital in scaling production.

The sections that follow lay out the strategic context, assess current programs against stated requirements, and close with recommendations addressed to Congress, the Department of Defense, and industry leadership. A companion appendix details methodology and sourcing.